

ERES Resonance Measurement Protocol

Instrument ID: ERES-RESONANCE-PROTOCOL-2026-001 **Version:** v1.0 **Layer:** GOOD (Data-Integrity) — Trinity handshake floor **Instance:** 01 of a polymorphic series — Environmental knob = **Sound** **Status:** Hypothesis-stage grounding study. Pre-registration required before data collection. **Framework:** ERES Institute for New Age Cybernetics · H2C2H · CCAL v2.1

0. Purpose

This protocol issues a single, narrow certificate: that a within-subject's **cross-channel physiological coherence** shifts measurably and repeatably in response to a defined change in environmental **sound**.

It is the GOOD-layer (Data-Integrity) handshake for resonance measurement — one concrete instantiation of the generic GOOD → SOUND → BEST Trinity interface. The **Environmental input** section (§3) is the only domain-specific part. Swapping it yields Instance 02 (light/circadian) and Instance 03 (controlled odorant) with everything else unchanged.

1. Claim under test

A single subject's cross-channel physiological coherence tracks a defined change in environmental sound, carrying information that no single channel carries alone.

Certificate issued on pass: *"This coherence reading is real and tracks the environmental input."* — Data-Integrity only. Nothing above this rung.

2. What this protocol does NOT claim

- It does **not** claim the coherence indexes a feeling.
- It does **not** claim it indexes a relationship, compatibility, or "who you can live with."
- It does **not** claim it indexes a personality type (Enneagram or otherwise).

Those are higher rungs (between-person synchrony; compatibility/type). Each requires its own chain of integrity beneath it. **The Infomediary qualifying any AnswerQuestion from this**

instrument may qualify only to the GOOD level — "integral reading" — and no further. Over-qualification is the leap this protocol exists to prevent.

3. Environmental input — the knob (*only domain-specific section*)

Parameter	Value
Variable	Ambient sound
Condition A	Calibrated silence (controlled low-dB baseline)
Condition B	Pink noise, ~60 dB SPL, fixed amplitude and spectrum
Measurement	Calibrated dB + spectrum log at the room (never inferred)

Rationale: pink noise is spectrally neutral to the ear and carries no semantic, musical, or valence content — it isolates *sound presence vs. silence*. Music is excluded: it smuggles in familiarity and emotional valence, the same uncontrolled-variable problem as using "feelings" as an input. Sound is the first instance because it is the most switchable knob — conditions alternate in seconds with no residue.

4. Subject channels — the body (four measured streams)

Channel	Measure	Sampling rate
Cardiac	HRV (chest strap, or validated PPG w/ R-peak detection)	1000 Hz
Respiration	Rate + depth (band, or ECG-derived)	25 Hz
Electrodermal	Skin conductance (EDA)	4 Hz (min)
Odor / VOC	Gas-sensor array, subject's local air	Continuous, native

Three channels is the minimum for "coherence" to mean more than a pairwise correlation. The fourth (VOC) is included because it is also one of the body's emission channels and earns its place in the resonance architecture.

- **Type:** Within-subject, repeated alternation.
 - **Baseline:** 5-minute seated rest before block 1 (settling period; standard).
 - **Blocks:** (A - B - A - B - A - B), six blocks, **5 minutes each**.
 - **Block length basis:** the short-term HRV standard (Task Force, 5 min) — set by the slowest-resolving channel; shorter windows make the HRV spectral estimate unreliable and weaken the coherence computation at its limiting channel.
 - **Total session:** 5 min baseline + 30 min blocks = **35 minutes**.
 - **Counterbalancing:** block order counterbalanced across sessions.
 - **Replication:** three full A/B pairs per session = within-session replication (the floor for SOUND-layer attestation — the result must repeat within the session before it earns clearance).
 - **Control:** one variable moves (sound); everything else held constant.
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6. Measurement — *this is the hypothesis*

Compute **cross-channel coherence** — the degree to which the four streams co-vary / phase-lock over each block. Established methods: cross-spectral coherence, phase-synchronization indices, cardiorespiratory coupling.

Compare the coherence pattern between Condition A and Condition B.

7. Falsification line — *this is what makes it a handshake, not a belief*

The study FAILS if cross-channel coherence does not separate Condition A from Condition B better than the single most informative channel does on its own.

If one channel (e.g. EDA) distinguishes silence from sound as well as the whole coherence pattern, there is no integrator — just one channel and noise. Passing **requires the joint pattern to carry information the best single channel does not**. The threshold and the comparison method are **pre-registered before data collection**.

Layer	Anchor	Gate (must confirm before next layer proceeds)
GOOD	Data-Integrity · Baidu	Channel signals intact, time-synced, artifact-checked. <i>No coherence computed until this confirms — first handshake.</i>
SOUND	Security-Clearance · SPRU	Coherence result attested: pre-registered threshold, counterbalancing, within-session replication — before it may mean anything.
BEST	One-Good · MIT	Result resolves into a named ERES quantity. RHC (Resonant Harmony Cycle) is the name BEST gives the within-person coherence result once the chain completes.

Dependency (build) order: Baidu → SPRU → MIT. **The order is the invariant. The contents are polymorphic.** A handshake whose steps could run in any order guarantees nothing — so the sequence is the contract, even though the rung contents instantiate differently across domains.

9. Polymorphism — how this becomes Instances 02 and 03

Replace §3 Environmental input only:

- **Instance 02:** light level / time-of-day (circadian).
- **Instance 03:** controlled odorant (present vs. absent).

Channels (§4), design (§5), measurement (§6), falsification line (§7), and handshake mapping (§8) are **unchanged**. One spec, three instantiations, climbed in order. Each pass strengthens the next.

10. Rungs above this one (*roadmap — not built here*)

1. **Rung 1 — within-person coherence** ← *this protocol*. Measurable now.
2. **Rung 2 — between-person synchrony.** Compare two subjects' coherence patterns. Real, studied; its own GOOD→SOUND→BEST chain.
3. **Rung 3 — compatibility / type.** "Who you can comfortably live with"; personality structure (test against validated trait models *first*, e.g. Big Five, before exploratory typologies). Hypothesized destination.

Each rung is earned only after the rung beneath it carries weight.

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